SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	**5**
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	**5**
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	**5**
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	**5**
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	**5**
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	**5**
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	**5**
8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	**5**

9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	**5**
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	**5**

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalis m
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

1. What is meant by Constraint?
A **constraint** is a rule applied to a table to maintain the **accuracy and integrity of data**.
Examples: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK.

2. What is the difference between Primary Key and Foreign Key?

Primary Key	Foreign Key
Uniquely identifies each record.	Connects two tables.
Cannot contain NULL values.	Can contain NULL values unless restricted.
Only one primary key per table.	A table can have multiple foreign keys.
Example: Student_ID	Example: Course_ID

Interview answer:
"A primary key uniquely identifies a record, whereas a foreign key is used to establish a relationship between two tables."

3. What is Normalization?
Normalization is the process of organizing data in a database to **reduce data redundancy and avoid data anomalies**.
The main normal forms are:
- **1NF** – Atomic values
- **2NF** – 1NF + no partial dependency
- **3NF** – 2NF + no transitive dependency

Simple answer:
"Normalization organizes tables properly and reduces duplicate data."

4. What is ACID Property?
ACID properties ensure that database transactions are **reliable and consistent**.
- **A – Atomicity:** Transaction happens completely or not at all.
- **C – Consistency:** Database remains valid before and after the transaction.
- **I – Isolation:** Transactions do not interfere with each other.
- **D – Durability:** Committed data is permanently saved.

Easy example: In a bank transfer, money should be deducted from one account and added to another completely.

5. How to connect a Java database using JDBC and ODBC?
JDBC stands for **Java Database Connectivity**. It is commonly used to connect Java applications to databases.
Basic JDBC steps:
1. Load the JDBC driver.
2. Establish a connection.
3. Create a statement.
4. Execute SQL query.
5. Process the result.
6. Close the connection.

Example:
```
Connection con = DriverManager.getConnection(
   "jdbc:mysql://localhost:3306/testdb",
   "root",
   "password"
);
```

ODBC stands for **Open Database Connectivity**. It is a standard database connectivity interface. Java can historically use ODBC through **JDBC-ODBC Bridge**, but modern Java applications normally use **JDBC directly with a database-specific driver**.

6. What is Collision?
In **hashing**, a collision occurs when **two different keys produce the same hash value/index**.
Example:
Hash function: key % 10

21 % 10 = 1
31 % 10 = 1
Both keys get index 1, so a **collision** occurs.

7. What are Collision Handling Techniques?
Collision handling techniques are methods used to store data when two keys get the same hash index.
The main techniques are:

1. **Separate Chaining**
2. **Open Addressing**
    - Linear Probing
    - Quadratic Probing
    - Double Hashing

8. What are the types of Collision?
The common collision-handling methods/types are:

- **Separate Chaining**
- **Linear Probing**
- **Quadratic Probing**
- **Double Hashing**

Easy interview answer:
"Collision itself occurs when two keys have the same hash value. We handle it using techniques such as chaining and open addressing."

9. Why should you use Collision Handling Techniques?
Collision handling is required because **multiple keys may map to the same hash location**.
It helps us:

- Store all records correctly.
- Avoid data loss.
- Retrieve records efficiently.
- Maintain good hash-table performance.

Interview answer:
"We use collision handling techniques to store multiple keys that map to the same hash index and to maintain efficient searching and insertion."

10. What is a Join?
A **JOIN** is used to **combine data from two or more tables** based on a related column.
Common types:

- **INNER JOIN** – Returns matching records.
- **LEFT JOIN** – Returns all records from the left table and matching records from the right.
- **RIGHT JOIN** – Returns all records from the right table and matching records from the left.
- **FULL OUTER JOIN** – Returns matching and non-matching records from both tables.

Simple example:

```
SELECT Student.name, Course.course_name
FROM Student
INNER JOIN Course
ON Student.course_id = Course.course_id;
```